

SAINT-YAN AP, France

WMO: 073790

Lat: 46.4083N

Lon: 4.0164E

Elev: 796

StdP: 14.28

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	19.4	23.3	14.1	11.6	22.7	19.1	14.8	26.3	23.0	50.2	20.2	49.0	3.0	60	0.487

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.6	91.4	68.9	87.1	68.0	83.4	66.8	71.1	85.2	69.6	83.3	68.2	80.6	6.3	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.8	101.7	76.3	65.4	96.6	74.2	64.0	91.8	72.3	35.4	85.5	34.1	83.6	32.8	80.9	77.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.5	16.4	14.1	12.0	96.3	10.6	3.7	4.3	99.0	-1.9	101.2	-7.8	103.3	-15.6	106.0
			11.7	74.4	10.1	2.0	4.5	75.9	-1.5	77.0	-7.1	78.1	-14.5	79.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	52.5	38.3	39.6	45.3	50.6	57.5	64.3	68.0	67.3	60.2	54.0	44.5	39.2
	DBStd	12.53	7.80	7.34	6.44	6.02	6.10	6.62	5.90	6.08	5.79	6.93	7.09	7.84
	HDD50	1480	366	296	171	65	9	0	0	0	2	40	190	341
	HDD65	4918	827	712	609	433	244	92	35	43	166	344	614	799
	CDD50	2382	4	3	26	82	241	430	557	535	307	164	26	7
	CDD65	347	0	0	0	0	11	72	127	113	21	3	0	0
	CDH74	4349	0	0	0	27	206	937	1496	1359	293	31	0	0
	CDH80	1618	0	0	0	1	40	339	591	577	68	2	0	0
Wind	WSAvg	5.2	5.5	6.2	6.1	5.8	5.1	4.9	5.0	4.4	4.1	4.5	5.3	5.7
Precipitation	PrecAvg	34.5	2.4	2.1	2.0	2.7	3.5	3.1	3.1	3.1	3.0	3.2	3.5	2.6
	PrecMax	41.7	4.9	4.0	5.7	6.2	7.4	7.8	6.5	5.8	6.1	6.8	7.5	5.2
	PrecMin	25.9	0.9	0.4	0.4	0.7	0.8	1.1	0.9	0.8	0.9	0.6	1.1	0.9
	PrecStd	4.4	1.1	1.0	1.1	1.4	1.5	1.6	1.4	1.3	1.5	1.3	1.4	1.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	58.2	61.9	70.2	77.6	84.5	94.7	96.2	98.3	86.7	78.0	66.3	60.2
		MCWB	51.6	51.2	55.9	60.7	66.0	70.1	68.6	69.3	65.5	63.8	57.8	52.4
	2%	DB	55.0	57.3	65.8	72.3	79.6	88.2	90.7	91.2	81.3	73.1	62.2	56.5
		MCWB	50.0	49.3	53.3	57.6	64.0	68.7	69.1	68.7	65.1	62.3	55.4	51.2
	5%	DB	52.4	54.0	61.5	68.5	75.5	83.5	86.6	86.4	77.1	69.6	58.8	53.9
		MCWB	48.0	47.5	51.4	55.9	61.9	67.2	68.1	68.0	64.0	60.7	54.0	49.9
	10%	DB	49.8	51.0	57.7	64.0	71.6	79.2	82.6	81.7	73.0	66.0	55.8	51.0
		MCWB	46.4	46.4	49.6	53.7	60.3	65.6	66.6	66.6	62.2	59.0	51.9	47.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.0	52.5	57.5	62.7	68.6	73.4	73.1	73.1	69.7	65.8	58.8	53.9
		MCDB	56.8	58.8	67.8	74.1	80.5	88.4	86.7	87.4	79.9	74.3	63.8	57.5
	2%	WB	50.5	50.1	54.2	59.3	65.8	70.8	70.9	70.7	67.0	63.4	56.4	51.9
		MCDB	54.1	55.4	62.7	70.8	76.5	84.1	85.2	85.5	77.5	70.9	60.7	55.5
	5%	WB	48.4	48.4	52.2	56.7	63.3	68.8	69.2	69.1	64.9	61.6	54.3	50.2
		MCDB	51.9	53.1	59.8	65.8	72.6	80.6	83.0	83.2	74.7	68.1	58.3	53.5
	10%	WB	46.5	46.6	50.4	54.5	61.1	66.7	67.6	67.5	63.1	59.5	52.1	48.1
		MCDB	49.5	50.6	56.9	62.3	69.7	76.8	80.2	79.5	71.1	64.8	55.3	51.0
Mean Daily Temperature Range	5% DB	MDBR	12.2	14.8	19.5	21.5	21.2	22.5	22.6	23.2	21.6	18.6	13.8	12.0
		MCDBR	15.9	21.4	27.8	30.2	29.4	30.9	30.9	32.9	29.4	24.1	18.3	15.4
		MCWBR	11.6	14.5	16.9	17.1	15.3	13.4	11.8	12.2	14.4	14.1	12.5	11.2
	5% WB	MCDBR	14.6	19.2	24.2	27.5	26.0	27.1	27.3	27.9	25.2	22.1	17.4	14.8
		MCWBR	11.2	13.5	15.4	16.0	14.4	12.8	11.6	11.9	13.0	13.8	12.5	11.2
Clear-Sky Solar Irradiance	taub	0.310	0.330	0.358	0.393	0.397	0.405	0.400	0.395	0.369	0.358	0.325	0.306	
	taud	2.461	2.403	2.337	2.244	2.284	2.300	2.308	2.335	2.382	2.413	2.459	2.463	
	Ebn at Noon	251	267	274	273	276	273	273	270	267	252	241	238	
	Edh at Noon	22	28	34	41	41	40	39	37	32	27	22	19	
All-Sky Solar Radiation	RadAvg	365	627	1032	1410	1650	1934	1870	1628	1255	781	424	305	
	RadStd	46	95	130	185	162	175	151	134	93	62	43	49	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (28 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page